# BP-Informed Diffusion Scores for Approximately Markovian Sequences

Calibrated grid BP, the message/model equivalence for single-Gaussian projections, rank-one non-Markov corrections, hybrid residual learning, and reverse dynamics

Thesis project continuation — second iteration

July 2026

**Abstract**

This report continues the study of belief-propagation (BP) computation of diffusion scores for sequence data with (approximately) Markov structure. Building on the previous note, we (i) audit and exactly reproduce the earlier package, identifying a numerical boundary-collapse artifact in its grid-projected Gaussian BP that contaminated all weak-noise results; (ii) prove that single-Gaussian moment-matched (assumed-density) BP on any *linear-transition* chain with Gaussian observation factors is identical to exact Gaussian BP under the covariance-matched Gaussian model, so that "Gaussian message error" and "Gaussian model error" coincide at this level; (iii) calibrate grid BP against the exactly solvable Gaussian AR(1) case, finding *spectral* accuracy with an abrupt small-$t$ failure driven by likelihood aliasing; (iv) quantify the corrected Gaussian-baseline error across heavy-tailed and bimodal innovation families; (v) show that for Gaussian AR(1)-plus-global-latent priors the non-Markov score correction is *exactly rank one* at every diffusion time, and that a neural network learning only this residual on top of a frozen Markov-BP score is an order of magnitude more sample- and parameter-efficient than a direct score network; and (vi) demonstrate in reverse-diffusion simulations that the Gaussian score reproduces second-order statistics while completely washing out heavy-tailed innovations, even though its trajectory-level effect is dynamically stable in the $L^2$ sense. Throughout we separate exact statements from numerical approximations.

## Contents

# 1 Setting and exact results

## 1.1 Model and forward convention

Clean data $a_{1:n} \in \mathbb{R}^n$ follow a Markov chain prior

$$p_0(a_{1:n}) = \mu(a_1) \prod_{i=2}^{n} K_i(a_i \mid a_{i-1}). \tag{1}$$

The forward noising process is the coordinatewise Ornstein–Uhlenbeck (variance-preserving) diffusion

$$\mathrm{d}X_t = -X_t\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}W_t, \qquad X_0 = a, \tag{2}$$

whose transition kernel gives, at time $t$,

$$x_i = \alpha_t a_i + \sqrt{\Delta_t}\, z_i, \qquad \alpha_t = e^{-t}, \quad \Delta_t = 1 - e^{-2t}, \quad z_i \sim \mathcal{N}(0,1) \text{ i.i.d.} \tag{3}$$

For unit-variance data, $\mathrm{Var}(x_i) = \alpha_t^2 + \Delta_t = 1$ at every $t$; all priors in this report are normalized to unit marginal variance so that $p_T \to \mathcal{N}(0, I)$. The sitewise likelihood is the exact Gaussian factor $\ell_i(a_i; x_i) = \mathcal{N}(x_i; \alpha_t a_i, \Delta_t)$.

## 1.2 Posterior chain, BP exactness, score identity

Because the prior factorizes over adjacent pairs and the likelihood factorizes over sites, the posterior is again a chain,

$$p_t(a \mid x) \propto \mu(a_1) \prod_{i=2}^{n} K_i(a_i \mid a_{i-1}) \prod_{i=1}^{n} \ell_i(a_i; x_i), \tag{4}$$

a tree-structured factor graph on which sum-product BP computes exact marginals $b_i(a_i) = p_t(a_i \mid x)$ after one forward and one backward sweep (full derivations in the previous report; they were checked line by line during the audit and are correct). Combined with the denoising (Tweedie) identity for the OU kernel,

$$\boxed{s_i(x,t) = \partial_{x_i} \log p_t(x) = -\frac{x_i - \alpha_t\, m_i(x,t)}{\Delta_t}, \qquad m_i(x,t) = \mathbb{E}[a_i \mid x_{1:n}],} \tag{5}$$

this yields the exact theorem: *for Markov-chain priors under coordinatewise OU noising, BP computes the exact diffusion score*, provided messages are represented exactly. The messages, however, are continuous functions; outside conjugate families they admit no finite parametrization. Everything that follows concerns the quantitative cost of representing them approximately, and what survives when the data are only *approximately* Markov.

## 1.3 The master error identity

If $\widehat{m}$ is any approximate posterior mean and $\widehat{s} = -(x - \alpha_t \widehat{m})/\Delta_t$, then for any reference pair $(m_{\mathrm{ref}}, s_{\mathrm{ref}})$ produced the same way,

$$\boxed{\widehat{s} - s_{\mathrm{ref}} = \frac{\alpha_t}{\Delta_t}(\widehat{m} - m_{\mathrm{ref}}), \qquad \frac{\alpha_t}{\Delta_t} \sim \frac{1}{2t} \quad (t \downarrow 0).} \tag{6}$$

All score errors are amplified posterior-mean errors, with amplification diverging at small noise. Every experiment in this package logs the numerical residual of (6) (column `identity_residual`);

it is at machine precision ($10^{-15}$–$10^{-12}$) throughout, as it must be, since it is algebra, not approximation. (When the compared scores agree to machine precision the *relative* residual is 0/0 and meaningless; the tables flag this case.)

Two further decompositions organize the study. Against an exact score,

$$\widehat{s} - s_{\text{true}} = \underbrace{(\widehat{s} - s_{\text{grid}})}_{\text{message/model approx.}} + \underbrace{(s_{\text{grid}} - s_{\text{true}})}_{\text{grid discretization}}, \tag{7}$$

and for approximately Markov data,

$$s_{\text{true}} = s_{\text{Markov}} + s_{\text{residual}}, \qquad s_{\text{hybrid},\theta} = s_{\text{BP},\mathrm{K}_\varphi} + r_\theta. \tag{8}$$

## 2 Audit of the previous package (Layer 1)

The previous package was rerun under its documented configuration (`-mode report -n-trials 12`, seed 11). All three CSV outputs reproduce *bit-for-bit up to machine precision* (largest deviation $2.1 \times 10^{-14}$, attributable to numpy/BLAS version differences). The mathematical derivations match the code. Two substantive findings change the interpretation of its results.

**F1: boundary collapse of grid-projected Gaussian messages.** The legacy `gaussian_projected_bp` evaluates each Gaussian message on the grid, applies the exact grid update, and moment-matches the result. At weakly informative times the outgoing message function is a near-flat ramp whose maximizer lies *outside* $[-A, A]$; moment-matching the truncated ramp yields a mean near the grid boundary and a spuriously small variance, and the next update pushes it further edge-ward (the backward integrand in $a$ peaks at $a'/\rho$, outside the grid) — a positive feedback loop. Diagnosed example (Gaussian prior, $t = 1.8$): backward messages collapse to $R_{\text{mean}} \approx -7.7$, $R_{\text{var}} \approx 0.08$ on a $[-8, 8]$ grid, producing a posterior-mean error of 6.9 at one site. This artifact—not Gaussian closure—explains the anomalous large-$t$ rows of the previous package (posterior-mean MSE $2.2 \pm 7.5$ at $t = 1.3$ on the Laplace chain, and MSE up to 5.0 in the *Gaussian* sanity check where projection should be exact). Figure 1 quantifies the artifact against the corrected implementation: it exceeds 6 in posterior-mean sup-norm for $t \geq 0.9$, precisely where the true closure error is below $10^{-2}$.

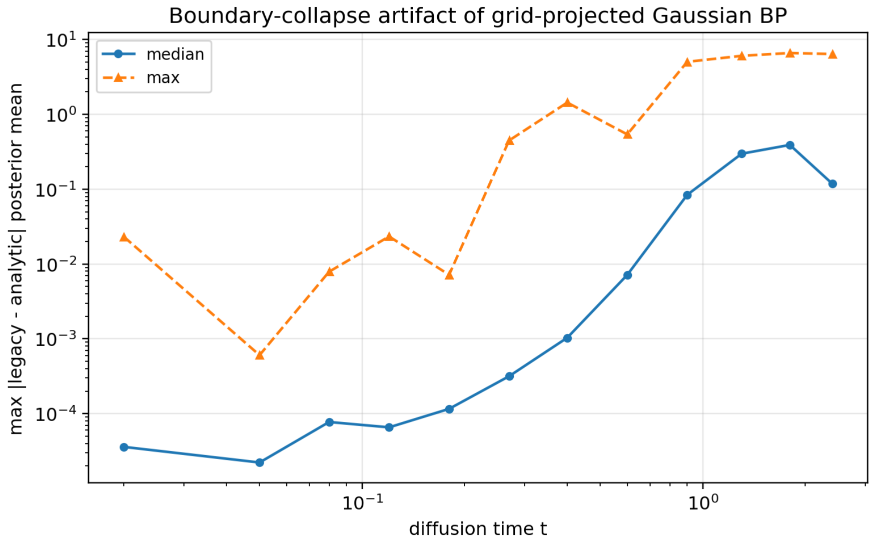


Figure 1: Audit finding F1. Sup-norm difference between the legacy grid-projected Gaussian BP posterior mean and the analytic information-form result on identical data (Laplace chain, $\rho = 0.85$). The two implement the same projection map, so the difference is purely numerical. The large-$t$ explosion is the boundary-collapse artifact.

**F2: message error = model error for single-Gaussian projections.** See Proposition 1 below: what the previous package measured as "Gaussian *message* approximation error" on the Laplace chain is mathematically the error of the covariance-matched Gaussian *model*. The numbers were right; the interpretation needs sharpening. The distinction between message-level and model-level approximation only becomes real for richer message families (mixtures, particles) or nonlinear transitions.

Minor findings: grid sizes were compared on different random draws (fixed by common random numbers keyed on $(\rho, t, \mathrm{trial})$); linear-domain likelihood rows can underflow entirely for small $t$ (fixed by row-shifted log-likelihoods, exact because per-site constants cancel); ad-hoc variance floors hid resolution failures (replaced by an explicit resolution check). The corrected package returns, in addition, the exact model evidence $\log p_t(x)$ from the BP normalization constants; it matches the closed-form Gaussian marginal likelihood in the calibration case (unit test).

# 3 Gaussian closure, and what a single Gaussian can ever measure

In the Gaussian AR(1) case messages close in information form $(\lambda, h)$: multiplication by the likelihood adds $(\alpha_t^2/\Delta_t,\ \alpha_t x_i/\Delta_t)$, and the linear transition maps moments $(m, v) \mapsto (\rho m,\ \rho^2 v + q)$. The corrected implementation (`gaussian_chain_bp`) performs exactly these updates analytically; a flat message is $\lambda = 0$ exactly, so no truncated moment-matching ever occurs.

**Proposition 1** (Single-Gaussian ADF-BP $\equiv$ matched Gaussian model BP)**.** *Let the clean chain have* linear *transitions* $a_i = \rho a_{i-1} + \varepsilon_i$ *with* $\mathbb{E}\varepsilon_i = 0$, $\mathrm{Var}(\varepsilon_i) = q$, *arbitrary innovation law, Gaussian initial density, and Gaussian sitewise likelihoods* (3)*. Then assumed-density BP that projects every message to a single Gaussian by moment matching produces exactly the messages, beliefs, posterior means, and score of* exact *BP under the Gaussian AR(1) prior with the same* $(\rho, q)$*.*

*Proof.* Forward pass: if the incoming message $L_i$ is Gaussian, the tilted product $L_i \cdot \ell_i$ is a product of Gaussians, hence exactly Gaussian $\mathcal{N}(\tilde{m}, \tilde{v})$ — projection is the identity here. The outgoing message is the law of $\rho\xi + \varepsilon$ with $\xi \sim \mathcal{N}(\tilde{m}, \tilde{v})$; whatever the innovation law, its first two moments are $(\rho\tilde{m},\ \rho^2\tilde{v}+q)$, so the moment-matched Gaussian is $\mathcal{N}(\rho\tilde{m}, \rho^2\tilde{v}+q)$ — identical to the Gaussian-innovation update. Backward pass: the outgoing function is $a \mapsto \mathbb{E}_\varepsilon[\tilde{g}(\rho a+\varepsilon)] = (\tilde{g} * \check{p}_\varepsilon)(\rho a)$ with $\tilde{g} = \mathcal{N}(\tilde{m}, \tilde{v})$; the convolution has mean $\tilde{m}$ and variance $\tilde{v} + q$ regardless of the innovation shape, so after the affine change of variable the moment-matched message is $\mathcal{N}(\tilde{m}/\rho, (\tilde{v}+q)/\rho^2)$, again the Gaussian-model update. By induction from the Gaussian initializations ($L_1 = \mu$ Gaussian, $R_n$ flat), all message parameters coincide; beliefs are products of Gaussians and coincide as well. □

**Remark 1.** *The projection discards the innovation law beyond its variance* at every step*, so nothing non-Gaussian ever enters. Consequently: (i) the honest implementation of the single-Gaussian baseline is the analytic information-form filter (no grid); (ii) measuring "message error" distinct from "model error" requires message families that can carry non-Gaussian information (Gaussian mixtures being the natural next step — with mixture incoming messages the tilted products are mixtures and multimodality survives the transition step); (iii) the proposition fails for* nonlinear *transitions, where the tilted forward moments genuinely depend on the innovation shape. This is verified numerically: the legacy grid projection agrees with the analytic filter to* $5 \times 10^{-3}$ *sup-norm at moderate* $t$ *on the Laplace chain (unit test), the residual being grid quadrature error.*

# 4 Layer 2: calibrating grid BP on the exactly solvable case

Grid BP represents messages by values on $M$ equispaced points in $[-A, A]$ with trapezoidal quadrature. Experiment 01 sweeps $M \in \{101, 201, 401, 801\}$, $A \in \{4, \ldots, 8\}$, $t \in [0.005, 2.4]$, $\rho \in \{0.2, 0.5, 0.8, 0.95\}$ against the exact Gaussian score, with identical random data across all grid configurations.

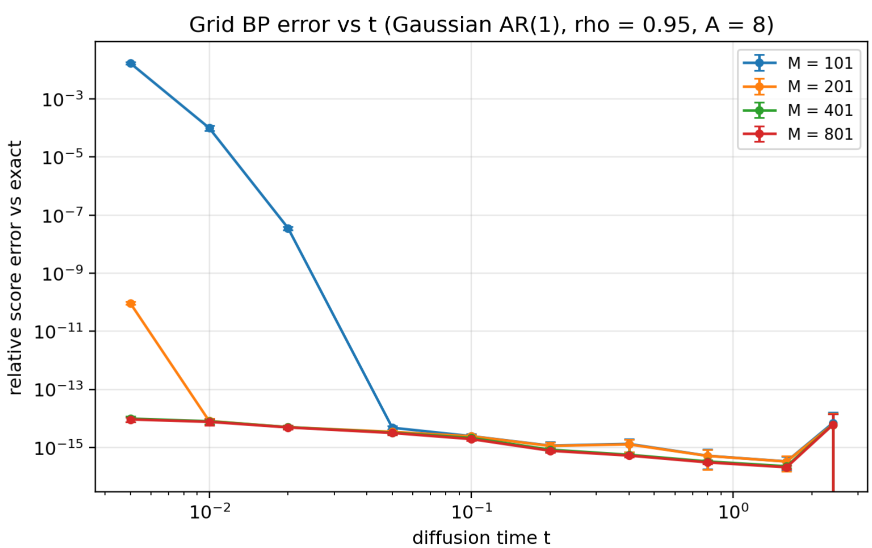


Figure 2: Relative score error of grid BP vs the exact precision-matrix score ($\rho = 0.95$, $A = 8$). The error sits at the $10^{-15}$ floor except where the likelihood width $\sqrt{\Delta_t}/\alpha_t \approx \sqrt{2t}$ approaches the grid step: failure is abrupt, not gradual.

**Spectral accuracy.** The trapezoid rule on analytic, rapidly decaying integrands converges exponentially in points-per-width (aliasing error $\sim e^{-2\pi^2\sigma^2/\mathrm{d}x^2}$), not at the naive $O(\mathrm{d}x^2)$ rate. Hence: at $M = 101$, $A = 8$ ($\mathrm{d}x = 0.16$) the error is $10^{-15}$ down to $t = 0.02$, degrades to $10^{-4}$ at $t = 0.01$, and fails ($1.7 \times 10^{-2}$) at $t = 0.005$; at $M = 201$ it is at the floor for all tested $t$. Tail truncation is visible only at $A = 4$ (error $10^{-6}$–$10^{-4}$ depending on $\rho$; at $A = 5$, $\leq 7 \times 10^{-6}$ for Gaussian tails). See Figures 3.

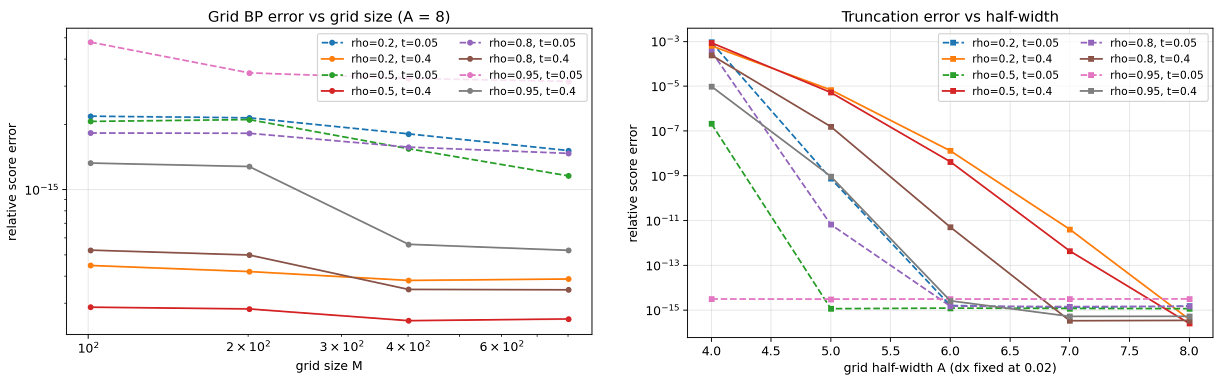


Figure 3: Left: error vs $M$ at $A = 8$. Right: error vs $A$ at fixed $\mathrm{d}x = 0.02$ (truncation isolated).

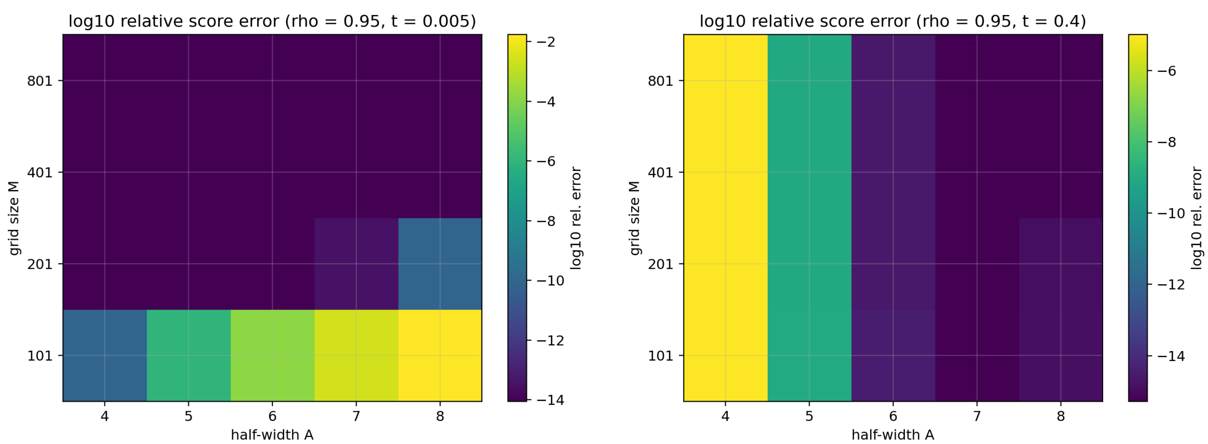


Figure 4: $(M, A)$ heatmaps of $\log_{10}$ relative score error at $\rho = 0.95$; $t = 0.005$ (left) shows the aliasing-driven failure region, $t = 0.4$ (right) shows the truncation-only regime.

**Recommended defaults.** The Gaussian calibration alone would admit $M = 101$, $A = 5$; because heavy-tailed innovations put more posterior mass in the tails, we adopt $M = 401$, $A = 8$ as the working reference for all non-Gaussian experiments (error floor in calibration, cost $O(nM^2)$ still trivial) and $M = 801$ for convergence checks. The binding small-$t$ constraint is $\sqrt{2t} \gtrsim 3\,\mathrm{d}x$, checked automatically (`resolution_ok`).

## 5 Layer 3: the corrected Gaussian baseline on non-Gaussian chains

### 5.1 Laplace chain, per-trial analysis (exp. 02)

On the Laplace chain ($\rho = 0.85$, $n = 40$, 60 trials per $t$, reference $M = 801$, $A = 8$), the corrected baseline gives a clean monotone picture (Figure 5): median relative score error 0.39 at $t = 0.02$, 0.17 at $t = 0.08$, $3.5 \times 10^{-2}$ at $t = 0.4$, $7.6 \times 10^{-5}$ at $t = 2.4$. The failure probability $P(\text{err} > 0.1)$ is 1 for $t \leq 0.08$ and 0 for $t \geq 0.9$. The grid self-convergence error ($M = 401$ vs 801) remains $10^4$–$10^7$ times smaller than the closure error at every $t$: the reference is trustworthy where it is used. The old package's large-$t$ anomalies are absent — they were artifact F1.

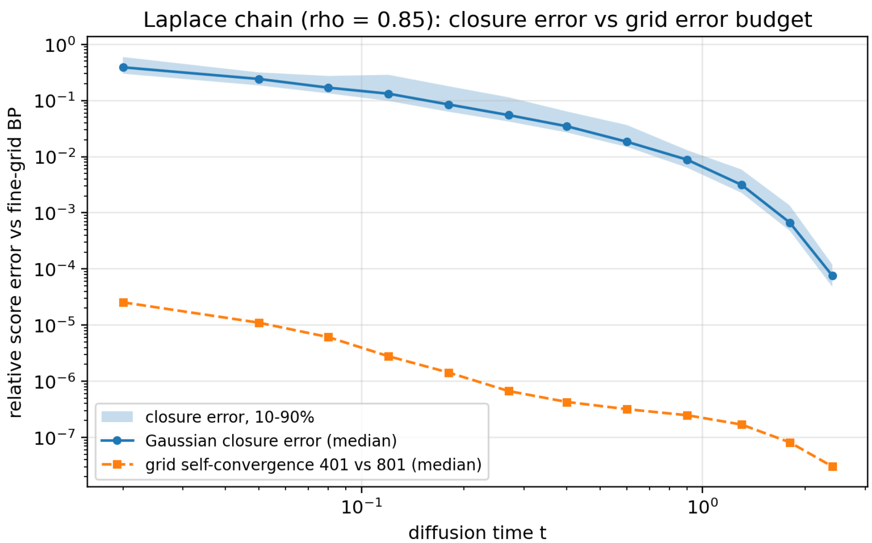


Figure 5: Gaussian closure error (median, 10–90% band) vs the grid error budget on identical data. Laplace chain, $\rho = 0.85$.

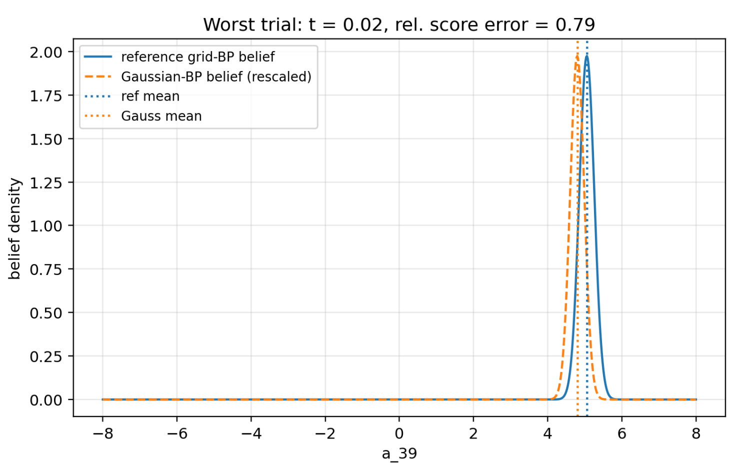


Figure 6: Worst trial across the sweep: the reference belief is sharply non-Gaussian and the single-Gaussian belief commits to a compromise; by (6) the mean shift is amplified by $\alpha_t/\Delta_t$.

### 5.2 Innovation sweep (exp. 03)

Six families with *identical* covariance $\rho^{|i-j|}$ and varying innovation shape: Laplace (excess kurtosis $+3$), Student-$t$ with $\nu = 5, 8$ ($+6$, $+1.5$), symmetric Gaussian mixtures with separation $\kappa = 0.3, 0.6, 0.9$ ($-0.18, -0.72, -1.62$; bimodal). Findings (Figure 7, full table in `outputs/exp_03.../SUMMARY_TABLE.r`

- **Noise level dominates**: every family's error decays to $\sim 10^{-3}$ or below by $t = 2.4$; at high noise the Gaussian likelihood dominates the posterior and closure is nearly free. All meaningful error lives at $t \lesssim 0.4$.
- **Both signs of non-Gaussianity hurt, bimodality hurts most**: at $t = 0.05$, $\rho = 0.5$: $\kappa = 0.9$ mixture 0.94, Laplace 0.52, $t_{\nu=5}$ 0.32, $\kappa = 0.3$ mixture 0.07. Strongly bimodal innovations are the worst case at small $t$, matching the mechanism of Figure 6.
- **Stronger correlation helps, not hurts**: at fixed $t$ the error *decreases* in $\rho$ (Laplace at $t = 0.05$: 0.43, 0.27, 0.13 for $\rho = 0.5, 0.85, 0.95$). Informative neighbours act like extra Gaussian observations and Gaussianize the local tilted posterior. The naive intuition that message errors "propagate farther" at large $\rho$ is not what happens on chains.

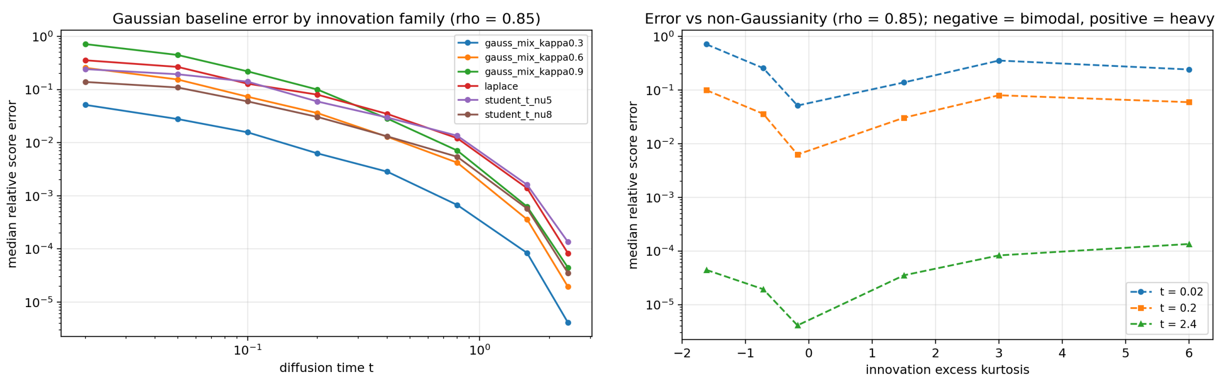


Figure 7: Left: median relative score error vs $t$ by family ($\rho = 0.85$). Right: error vs innovation excess kurtosis (negative = bimodal, positive = heavy-tailed) at three times.

By Proposition 1 these numbers are simultaneously the single-Gaussian *message* error and the matched Gaussian *model* error. Separating the two requires mixture messages; this is the clearest next algorithmic step and is left for the next iteration.

## 6 Layer 4: approximate Markovianity

Here the prior is *not* a chain, but Gaussian, so every score is exact linear algebra and the pure "Markov approximation error" is isolated with no grid and no Monte Carlo in the operator analysis.

### 6.1 Model A: AR(1) plus global latent — the residual is rank one

Let $a = (y + \beta g)/\sqrt{1+\beta^2}$ with $y$ a Gaussian AR(1)($\rho$) chain and $g \sim \mathcal{N}(0,1)$ global, so $\Sigma_0 = (\Sigma_y + \beta^2 \mathbf{1}\mathbf{1}^\top)/(1+\beta^2)$.

**Proposition 2** (Rank-one non-Markov correction)**.** *Let $S_t = \alpha_t^2 \Sigma_{\text{chain}} + \Delta_t I$ be the noisy covariance of any chain prior and let the true noisy covariance be $\Sigma_t = S_t + c\,\mathbf{1}\mathbf{1}^\top$ with $c = \alpha_t^2 \beta^2/(1+\beta^2)$. Then by the Sherman–Morrison identity*

$$s_{\text{true}}(x) - s_{\text{chain}}(x) = \frac{c\,(\mathbf{1}^\top S_t^{-1} x)}{1 + c\,\mathbf{1}^\top S_t^{-1}\mathbf{1}}\; S_t^{-1}\mathbf{1}, \tag{9}$$

*i.e. the non-Markov score correction is exactly rank one at every $t$: a fixed smooth direction $S_t^{-1}\mathbf{1}$ with an $x$-linear scalar coefficient.*

Numerically (exp. 04): the residual operator $D_t = \Sigma_{\text{chain},t}^{-1} - \Sigma_t^{-1}$ of the naive chain has $\sigma_2/\sigma_1 \sim 10^{-12}$ (exact rank one), and chain-BP + correction (9) agrees with the exact score to $5 \times 10^{-14}$ — this *is* augmented BP with the global latent integrated out in closed form. The best-matched chain (Chow–Liu, adjacent pairwise marginals matched) has smaller residual *norm* (e.g. 0.066 vs 0.086 median relative residual at $\beta = 1$, $t = 0.2$) but spreads it over many directions: structure and norm trade off. Residual magnitudes are graceful in $\beta$: median relative score residual at $t = 0.2$ of 0.002/0.011/0.03/0.07 for $\beta = 0.1/0.25/0.5/1.0$ (Chow–Liu).

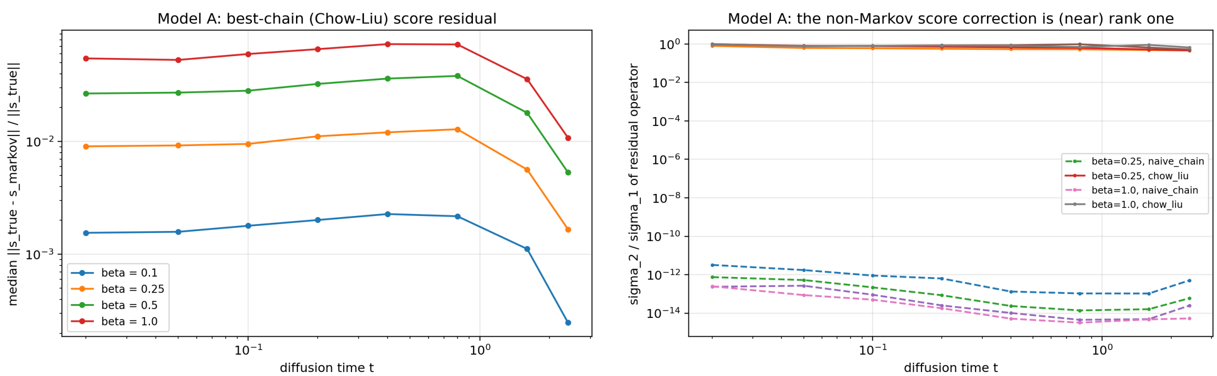


Figure 8: Model A. Left: relative score residual of the best chain approximation. Right: $\sigma_2/\sigma_1$ of the residual operator — the naive chain's correction is exactly rank one (dashed, at numerical zero); Chow–Liu trades structure for norm.

## 6.2 Model B: weak long-range precision coupling

With $Q = Q_{\text{AR1}} + \gamma Q_{\text{long}}$ (exponentially decaying long-range couplings, diagonally compensated, covariance renormalized to unit diagonal), the Chow–Liu chain residual grows like 0.084/0.13/0.17/0.23 (median, $t = 0.2$) for $\gamma = 0.05/0.1/0.2/0.4$, with effective rank $\approx 30$: a genuinely high-rank correction. Low-rank latent structure (Model A) is the favourable regime for Markov-plus-residual decompositions; generic long-range couplings are not, although the residual *norm* remains modest for weak $\gamma$.

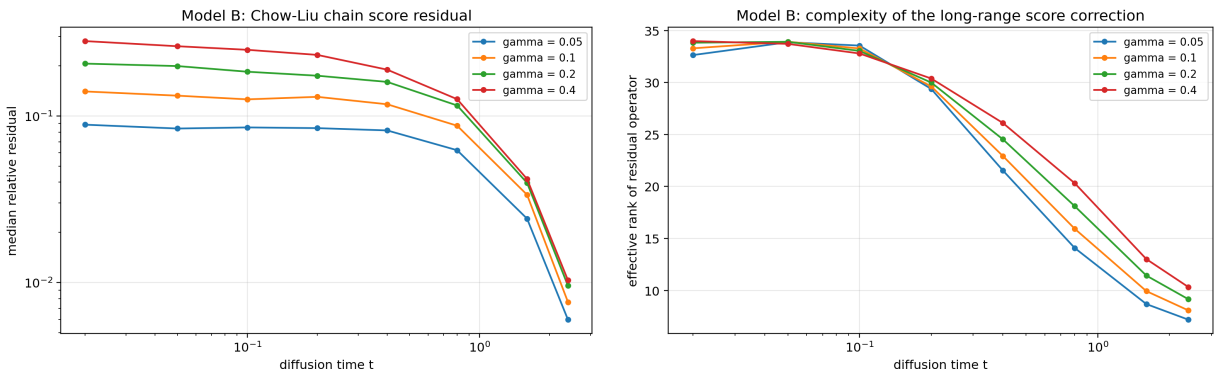


Figure 9: Model B: residual norm (left) and effective rank of the residual operator (right).

### 6.3 Hybrid learning: BP baseline plus neural residual

On Model A ($\beta = 0.5$, $\rho = 0.8$, $n = 40$), supervised regression on exact-score targets, times log-uniform in $[0.02, 3]$: a direct MLP $s_\theta(x, t)$ versus the frozen Chow–Liu Markov-BP score plus an identical-architecture MLP learning only the residual (8). Test relative score error on 2048 held-out samples:

| $N_{\text{train}}$ | 64 | 256 | 1024 | 4096 |
|---|---|---|---|---|
| direct MLP | 0.88 | 0.75 | 0.29 | 0.20 |
| hybrid (BP + residual MLP) | **0.23** | **0.060** | **0.015** | **0.014** |
| Markov BP alone (no learning) | | 0.031 | | |

The hybrid improves on the free Markov baseline from 256 samples on and is $\sim 20\times$ better than the direct network at every size; the direct network does not reach the *no-learning* baseline even at 4096 samples. Parameter efficiency is equally lopsided: a width-16 residual net (1.7k parameters, error 0.017) outperforms a width-256 direct net (87k parameters, error 0.33).

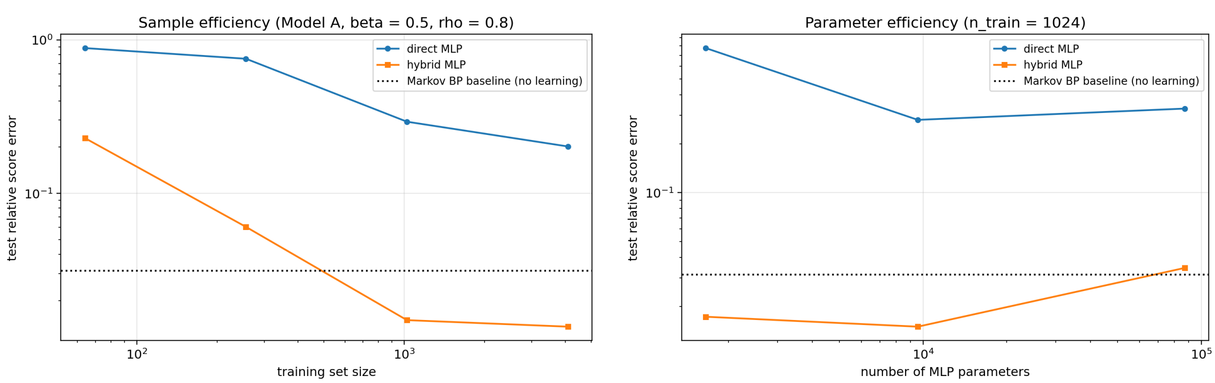


Figure 10: Sample (left) and parameter (right) efficiency of direct vs hybrid score learning on Model A.

**Remark 2** (Honest caveats)**.** *The Markov model is* known *here, not estimated ($K_\varphi$ given, not learned); the comparison isolates the value of the structural baseline, not of a full estimation* *pipeline. Targets are exact scores (supervised), not denoising score matching; the sample-efficiency gap could change under DSM noise. And Model A is the friendly case: its residual is exactly rank one and $x$-linear — which is precisely the point: when approximate Markovianity holds with low-rank corrections, the residual is a* much *easier learning problem than the full score.*

# 7 Reverse-diffusion dynamics (exp. 05)

Convention: reverse SDE $x \leftarrow x + h\,(x + 2s(x,t)) + \sqrt{2h}\,\xi$ integrated on a geometric grid from $T = 3$ to $t_{\min} = 0.01$ (80 steps, 200 trajectories); probability-flow ODE $\dot{x} = -x - s$ with Heun steps for reconstruction. Paired comparisons share initial noise and Brownian increments.

## 7.1 Does the Gaussian score wash out non-Gaussian structure?

Generating from the Laplace chain ($\rho = 0.85$) with the grid-BP score versus the Gaussian-BP score:

| | innovation exc. kurtosis | marginal var | lag-1 corr. |
|---|---|---|---|
| clean prior samples | 2.72 | 0.94 | 0.837 |
| reverse, grid-BP score | **2.86** | 1.06 | 0.828 |
| reverse, Gaussian-BP score | **0.12** | 1.05 | 0.832 |

Both samplers reproduce the second-order statistics; only the grid-BP score recovers the heavy-tailed innovations. This is the dynamical face of Proposition 1: the Gaussian score *is* the score of a Gaussian model, so its reverse diffusion generates (asymptotically) Gaussian samples — the non-Gaussianity is not distorted but *erased*.

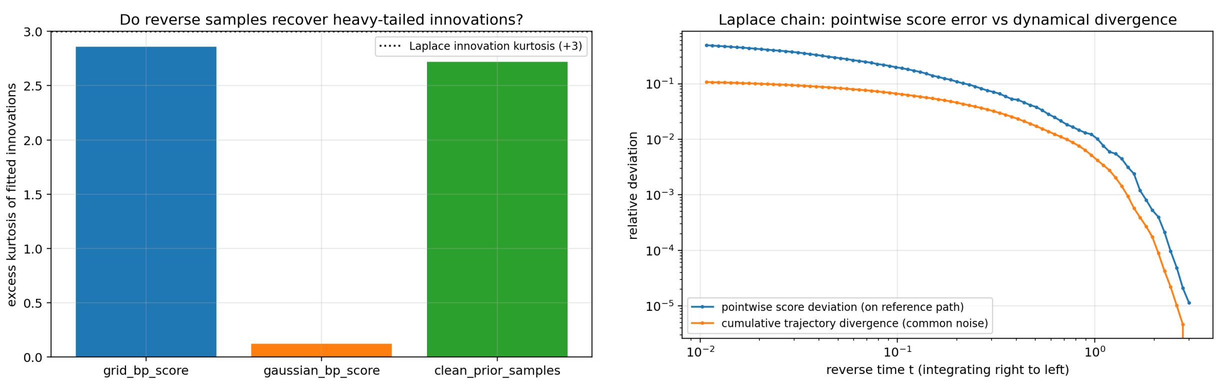


Figure 11: Left: innovation kurtosis of reverse samples. Right: pointwise score deviation on the reference path vs cumulative common-noise trajectory divergence.

## 7.2 Pointwise error vs dynamical error

Along the reverse trajectory, the pointwise Gaussian-vs-grid score deviation is $\sim 10^{-5}$ at $t = 3$, crosses $10^{-1}$ near $t \approx 0.15$, and reaches 0.49 at $t_{\min}$; yet the cumulative trajectory divergence under common noise saturates around 0.11. Two conclusions. First, the dynamics are *stable*: score error is not amplified along the flow (the late-time contraction of the reverse SDE limits accumulation; errors arrive mostly after the trajectory's large-scale structure is frozen). Second, stability in $L^2$ is deceptive: the modest terminal displacement is exactly what carries the higher-moment structure, as the kurtosis table shows. Small pointwise score error at moderate $t$, plus stability, is *not* sufficient for distributional correctness in the tails.

Reconstruction from $t = 1$ (probability-flow ODE) shows the two scores recovering correlation with the clean sample nearly identically (second-order task, consistent with the above); see `setup2_correlation_recovery.png`.

### 7.3 Model A: recovering the global mode

Generating from Model A ($\beta = 0.5$) with four scores: exact, Chow–Liu Markov, naive Markov, and hybrid (chain + rank-one). Top eigenvalue of the terminal sample covariance (true value 12.7; finite-batch/discretization biases affect all samplers equally — exact-score value 10.5): exact 10.5, hybrid 10.5 (bitwise-identical trajectories, as (9) is exact), Chow–Liu 8.1, naive 5.6. A Markov score cannot generate the global collective mode; the rank-one correction restores it exactly.

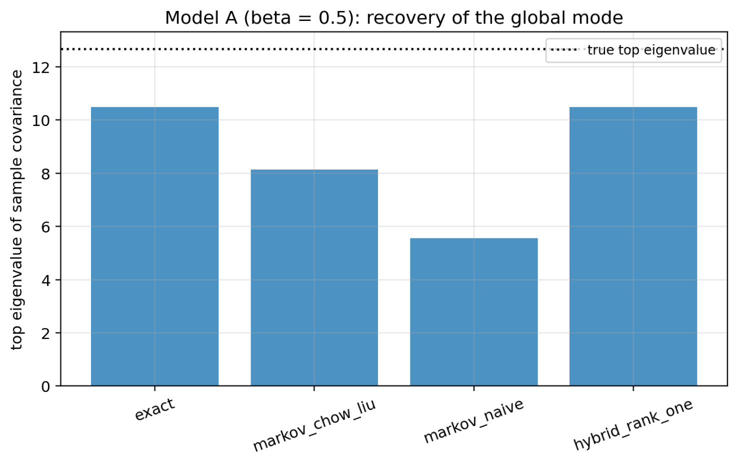


Figure 12: Model A: top eigenvalue of the sample covariance of reverse-generated samples.

## 8 Interpretation, limitations, and next steps

**What is exactly established.** (i) For Markov-chain priors under coordinatewise OU noising, BP computes the exact score; the entire computational question is message representation. (ii) Single-Gaussian projected BP on linear chains equals matched-Gaussian-model BP (Prop. 1); its failure modes are therefore model failures: erased tails and modes, worst at small $t$ by (6). (iii) For chain-plus-low-rank-latent Gaussian priors, the non-Markov score correction is exactly low-rank (Prop. 2); augmented BP is closed-form.

**What is numerically established.** Grid BP is spectrally accurate and abruptly fails only below a predictable $t$; with $M = 401$, $A = 8$ it is a sound reference for every experiment here. The corrected Gaussian baseline degrades smoothly and predictably in $(t, \text{kurtosis}, \rho)$. Hybrid BP-plus-residual learning dominates direct score learning by an order of magnitude on a model with genuine low-rank non-Markovianity. Reverse dynamics under approximate scores are $L^2$-stable but distributionally wrong precisely in the structure the approximation discards.

**What this does not show.** BP does not solve general diffusion models: real priors are unknown, non-chain, and high-dimensional; our non-Gaussian ground truths are numerical (grid), not analytic; the hybrid experiments assume the Markov model known and use supervised score targets. Claims are limited to: *if the clean distribution is (approximately) a Markov chain, BP-based scores are exact (resp. structured baselines), and the non-Markov residual can be dramatically simpler than the full score — in the low-rank-latent case, exactly rank one.*

**Next steps, in order of leverage.**

1. **Gaussian-mixture messages**: the first family where message error $\neq$ model error; measure how many components close the small-$t$ gap on the bimodal chains (the worst case found here).
2. **Estimated** $K_\varphi$: learn the transition from data (parametric or nonparametric), quantify how transition-estimation error propagates through (6) to the score, completing the hybrid pipeline $s_{\mathrm{BP},\mathrm{K}_\varphi} + r_\theta$ end to end.
3. **Denoising-score-matching training** of the residual (no exact targets), to test whether the sample-efficiency gap survives DSM variance.
4. **Model C** (non-Gaussian chain + global latent): Rao–Blackwellized BP with a quadrature over the latent times grid-BP chains — machinery already in place; the reference is small-$n$ quadrature or particle methods.
5. **Nonlinear transitions**, where Proposition 1 fails and moment-matched messages genuinely differ from any Gaussian model — the sharpest place to study message-level approximation per se.

# A Package layout and reproduction

| Path | Content |
|---|---|
| `src/` | library: priors, noising, grid BP (+ evidence), information-form Gaussian BP, exact scores, Markov approximations, numpy MLP, samplers |
| `experiments/exp_01..05` | all experiments; `-quick` smoke mode; CSV/JSON/PNG outputs |
| `outputs/` | committed results of the full runs used in this report |
| `notebooks/01..04` | executed analysis notebooks |
| `tests/` | 14 pytest checks (exactness, identities, equivalence, convergence) |
| `audit/AUDIT_NOTE.md` | full Layer-1 audit |

Determinism: all randomness flows through `src.utils.rng_for` (seeded by experiment/setting keys), so method comparisons are paired by construction. Provenance (package versions, timestamps) is written into every `params.json`.

# B Relation to the previous report

The general theorem, its proof, and the Gaussian-closure algebra are unchanged from *From Gaussian AR(1) Belief Propagation to General Markov-Chain Diffusion Scores* (July 2026) and are not repeated in full here. The Laplace-chain numbers of that report remain valid at $t \leq 0.9$ and are superseded at $t \geq 1.3$ by the corrected baseline (artifact F1). Its interpretation of "Gaussian message error" should be read as "matched Gaussian model error" per Proposition 1.

## References


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[5] G. Biroli and M. Mézard, *Generative diffusion in very large dimensions*, arXiv:2306.03518, 2023.

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